Inference with Normal Distribution

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MATH 240

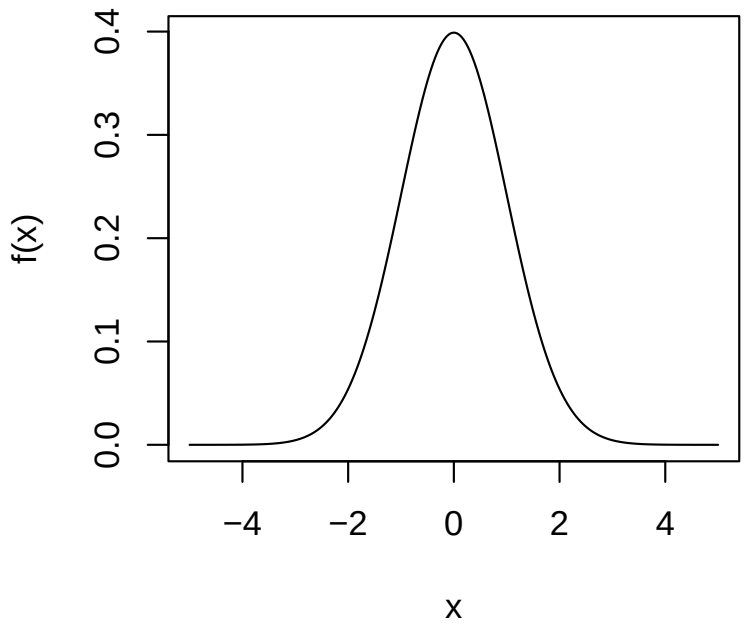
Week 10, Lecture #3 — Ch. 13

Reminders

- **HW7** due on Monday at 11:59pm
- **Exam #2** on next Friday during lecture
  - Two pages of notes (8.5 by 11", front and back)
  - Bring a calculator and pencil
  - Logistic regression, probability, hypothesis testing, confidence intervals, normal distribution, and decision errors

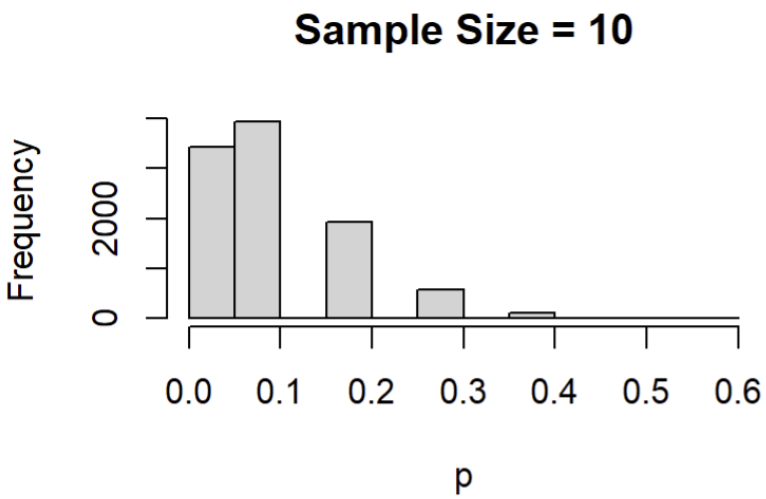
Use of Mathematical Models

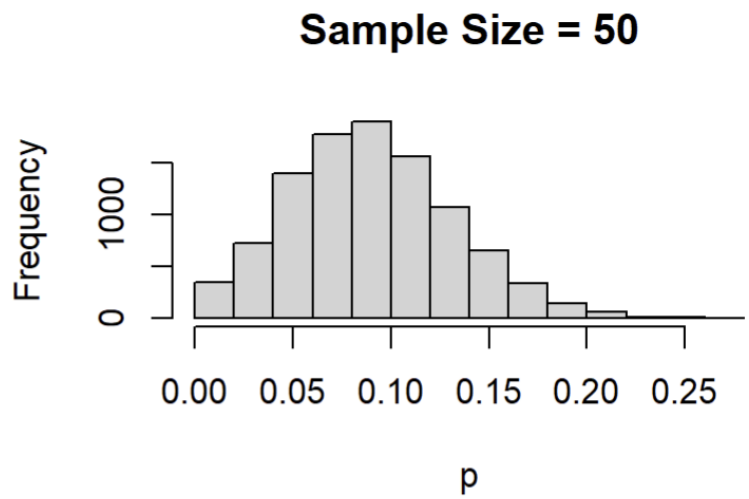
So far, we've explored hypothesis testing through the use of computational techniques. We can also use models in order to conduct a test, mainly through the use of the Normal distribution.

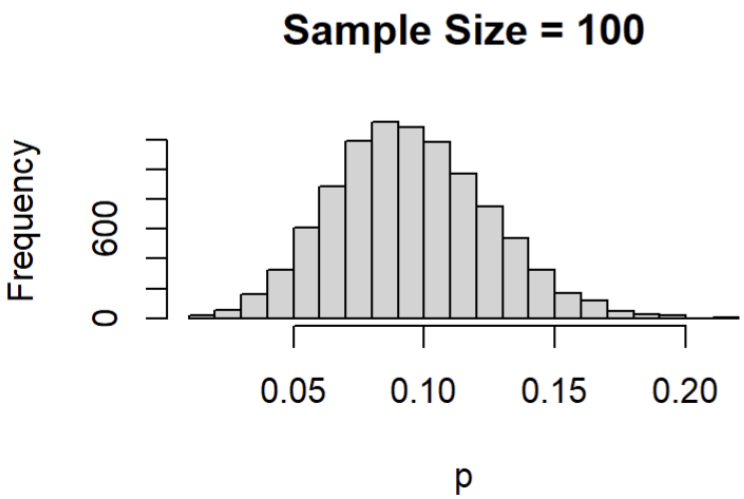


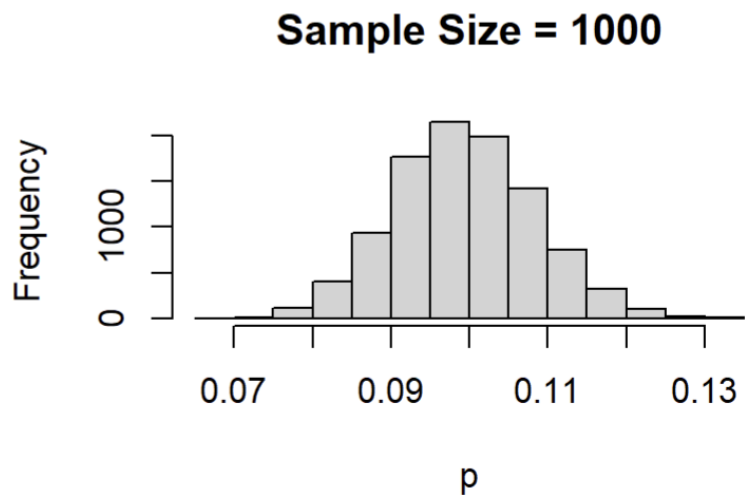
Objective: Use the Normal distribution on sample data to find probabilities and quantiles, which will later be seen in hypothesis testing.

Sampling Distributions









Aside: Continuous Random Variables

A continuous random variable X is a numeric summary of a random experiment that can take on any value in an interval.

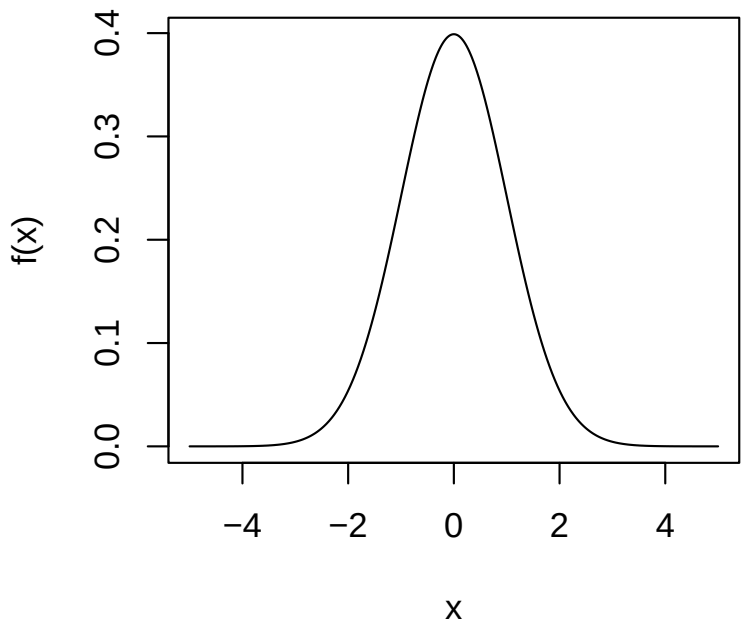
- e.g. box office gross for a randomly selected movie
- e.g. birth weights for a randomly selected infant
- e.g. time in between orders at a fast-food restaurant

For a continuous random variable, we can think of its probability mass as being smeared over the entire support, forming a smooth density curve.

Probability Density Function

The probability density function is a function $f_X(x)$ that is:

1. Non-negative (e.g. $f_X(x) > 0$)
2. Has a total area under the curve of one (e.g. $\int_{-\infty}^{\infty} f_X(x)dx = 1$)



Continuous Random Variables

- Point probabilities $P(X = x)$ are always 0 for a continuous random variable
- If we want to find probability between a and b, we can find:

$$P(a < X < b) = \int_a^b f_X(x)dx$$

- For a lower-tailed probability, we can use the **cumulative distribution function**

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(t)dt$$

Normal Distribution

The Normal distribution describes a continuous random variable whose:

- Density function $f_X(x)$ is symmetric, unimodal, and bell-shaped
- Provides functionality for the empirical rule, z-scores, and the central limit theorem
- Denoted as:

$$X \sim N(\mu, \sigma^2)$$

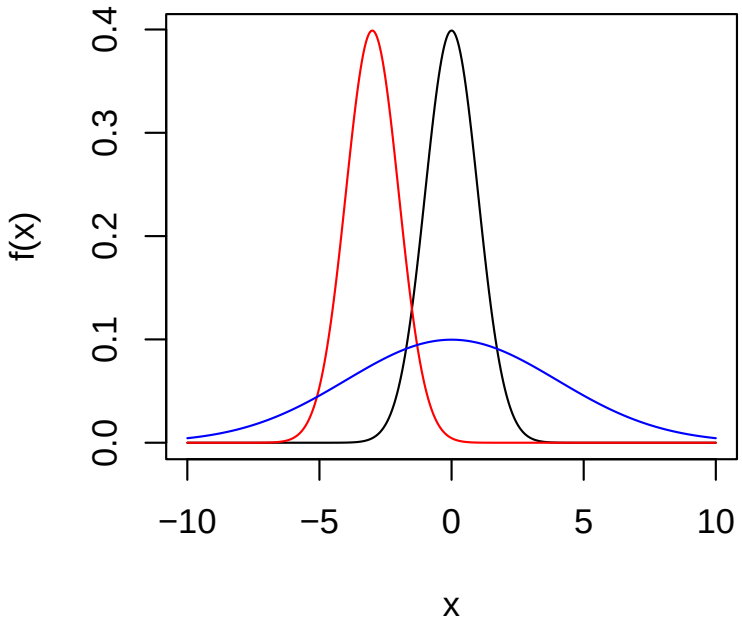
  - μ: population parameter for mean
  - σ^2: population parameter for variance
- $Z \sim N(0, 1)$ represents a standard normal random variable

Normal Distribution

- Distribution function:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

  - For $-\infty < x < \infty$
- Different curves:
  - $X \sim N(\mu = 0, \sigma^2 = 1)$
  - $X \sim N(\mu = -3, \sigma^2 = 1)$
  - $X \sim N(\mu = 0, \sigma^2 = 4)$

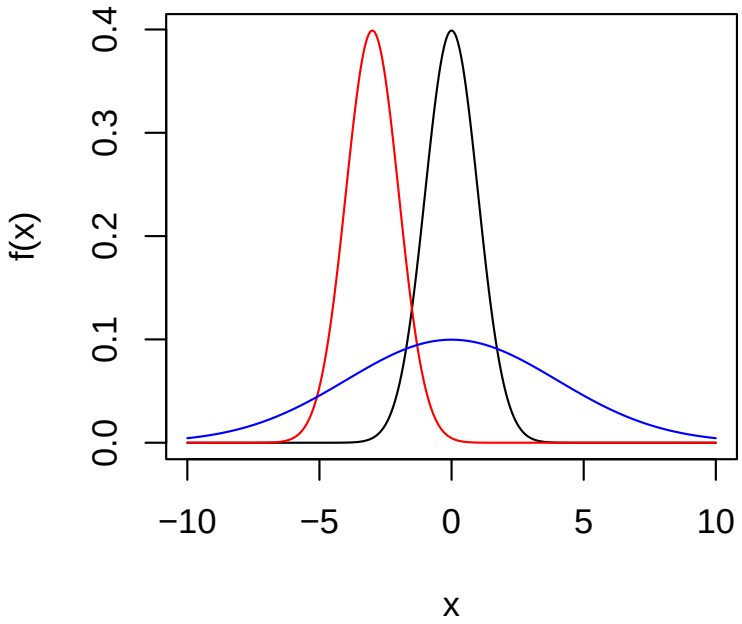


Normal Distribution

- To find $P(a < X < b)$, we would typically use integration on $f_X(x)$:

$$\int_a^b f_X(x)\,dx$$

- No closed-form solution – this is where the empirical rule and z-scores come into play!

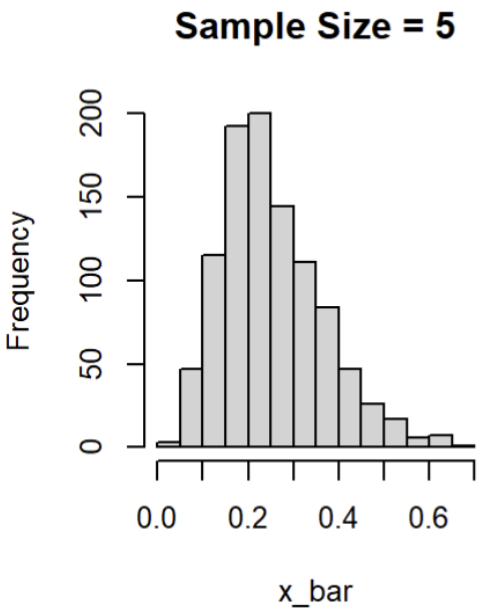


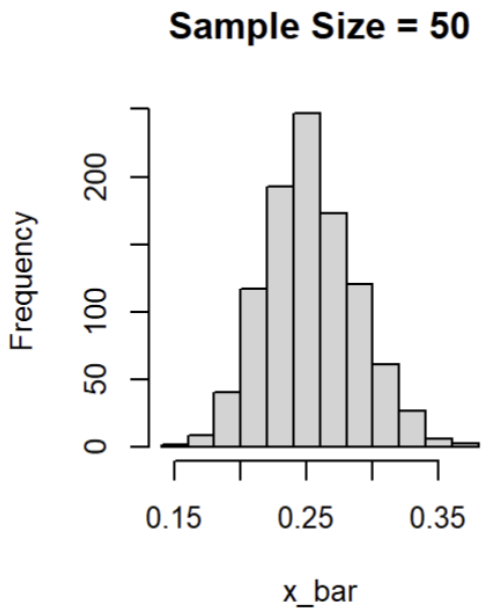
Central Limit Theorem

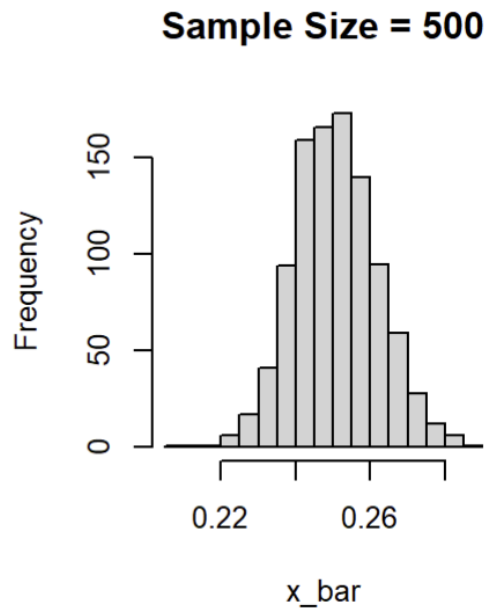
The **Central Limit Theorem** is one of the most foundational theorems in statistics.

- Provides an alternate way of approximating the sampling distribution for statistics such as $\bar{x}$ and $\hat{p}$
- Theorem: Sampling distribution of one of these two statistics can be approximated by a normal distribution, provided that:
  1. The observation in the sample are **independent**
  2. The sample size n is large
- The approximation is:
  - $\bar{X} \sim N(\mu, \sigma^2/n)$
  - $\hat{p} \sim N\left(p, p\left(1-p\right)/n\right)$

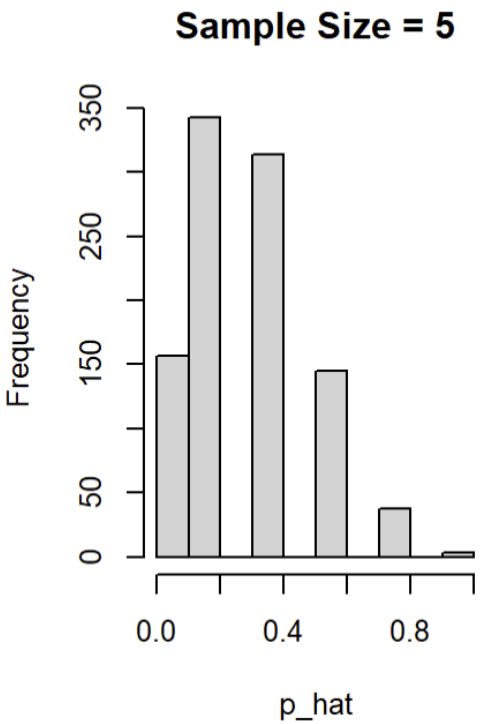
Central Limit Theorem

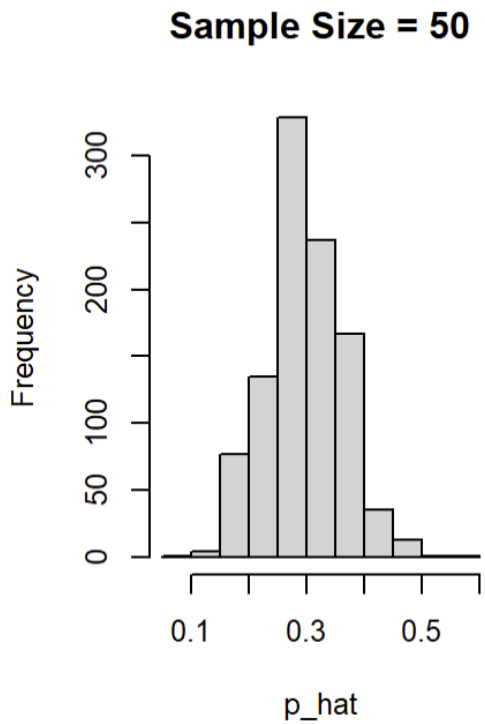


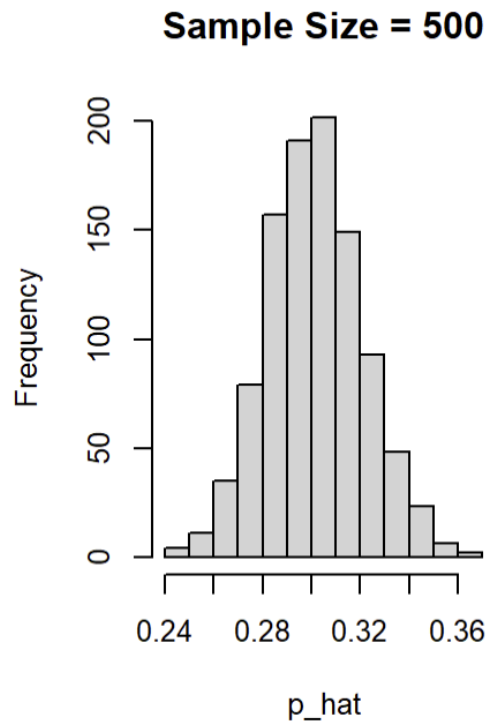




Central Limit Theorem







Connection between CLT and Hypothesis Testing

If I can approximate my sampling distribution with a Normal distribution...

- Then we can use its properties to find the p-value and parts of the confidence interval for our test statistic
- The test statistic can now be converted to some value for a (standard) normal random variable, and we can then find the area under the curve to find its p-value
- So back to an older question: how can we find probabilities for a normal random variable?

The Use of Z-Scores

Making comparisons between two Normal random variables of different scales and shapes can be troublesome:

- e.g. test scores of one exam out of 50 points vs. another out of 75 points

z-scores provide us with an opportunity to make this comparison.

1. Standardize and scale the value from the Normal random variable

$$z = \frac{x - \mu}{\sigma}$$

2. Find probabilities and quantiles based on the new values

Finding Z-Scores

College admissions wants to understand the performance for someone with a 1300 on the SAT vs. 27 on the ACT. Suppose we know that:

- mean SAT score: 1020; standard deviation: 194
- mean ACT score: 21; standard deviation: 5.4

Exercise: Compute the z-scores to see who performed better.